

Maintaining agency in AI-generated works of art and design

Deliberate creative processes

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ABSTRACT

This paper responds to the unfolding debate on the use of generative AI tools in art and design. As the conversations and technological frameworks remain in their early stages the paper focuses on emerging possibilities and methodologies rather than on broader claims or conclusions. The discussed methods can serve as an effective entry point for education and curricula developments with practical advice and conceptual framing to contextualize emerging AI tools in a broader artistic culture and legacy. While the discussion on autonomous creativity is a significant part of the future of AI system developments and their relationship with humanity, this paper specifically limits its contribution to human-to-AI interactions as an important step in understanding intentionality, agency, and identity in that newly formed landscape.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; Interaction design; Interaction design theory, concepts and paradigms.

KEYWORDS

Generative AI, Stable Diffusion, AI-based creative process, Artificial intelligence

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1 INTRODUCTION

Artificial intelligence (AI) is a dynamically evolving technology impacting creative professionals and the future of academic education [Newton, 2021] [As and Basu, 2021]. The relationship between AI technology and human creativity—collaboration, dependency, or interdependency—will shape the discourse around AI in the near future [Newton, 2019] [Huang et al., 2021]. This warrants, and even demands, the development of strategies for creative collaboration and co-authoring with AI.

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This paper looks at AI-generative processes from the artistic tradition of working with the medium and careful control of tools (parameters) involved in creativity. Specifically, the paper discusses emerging concepts and practices as important frameworks defining AI-based design and art curricula. The focus on methodology is necessitated by the significant shift in the way AI tools reframe conceptualization and expression in arts and design. While the ability to express an intent or concept is an important part of artistic outcomes, creative processes and methods are imperative in developing and enabling creative skills in students and young artists. This paper reflects the current state of generative AI rather than what AI creativity may be in the future. However, there is value in investigating creative processes utilizing AI even if current technologies are transitory (evolving) as they still help to define the relationship between human creativity and the one based on AI technologies.

Novelty does not always mean innovative and creative thinking as being different is not synonymous with qualitative enhancement. The questions of authorship, differentiation and creative leap are even more acute with AI-based creativity than it was in the past traditions and practices. One of the determining factors could be whether the work is derivative of what is already around or it is transformational. This argument is often used in the intellectual property context and should be applicable here. A similar distinction has been made as early as the mid-16th century by Giorgio Vasari [Vasari, 2007], who identified two aspects of artistic work: imitation (*imitazione*) and invention (*invenzione*) with their often mutually informing and dialectic relationship. This paper tries to touch on the mechanics of this process by considering the utility of both types of outputs—derivative (*imitazione*) and transformational (*invenzione*)—with each aspect serving a valid creative purpose: the former as a class of visual objects that maintain an overall aesthetic cohesion reinforcing style consistency and the latter as a creative tool that transcends the precedent.

The ability of AI-generative technologies to create deeply aesthetically-satisfying work alongside generic artwork raises the question of validation—distinguishing between high quality from mediocre outputs. Evaluation is significantly different from generation and currently, it is left to humans to set this baseline. This is where the intentionality and agency of AI-generated art reside.

2 INTENTIONALITY AND AGENCY

The question often raised in the context of AI creativity refers to the source and the direction of intentionality and agency in the creative process [Terzidis et al., 2023]. Who is the ultimate creator? At what point imagery generated by AI technologies can be granted full authorship autonomy? While the debate is ongoing, this paper emphasizes the role artists and designers can play in AI-enhanced

creativity pointing to emerging practices and methodologies. By accepting this human-driven process, it is understood that intentionality and agency remain with human artists, who also serve as exclusive validators of produced artwork. This does not preclude future developments where the center of gravity in this debate could, or most likely will, change toward generative AI systems. However, human-AI interactions and creativity remain the next logical step in the evolution of AI technologies.

Questions about intentionality and agency have been raised before. Surrealist works are defined by a concept, process, often unusual techniques, and the use of materials. Involuntary sculptures or exquisite corpses serve as immediate examples where the authorship is unconscious or collective and distributed. Similarly, the works of Jackson Pollock depended on the uncertainty of tools and processes as well as human kinetic motion with no chance to repeat or refine the outcomes. AI creativity feels and behaves in similar ways, where human authorship and agency begin to blur with tools and processes.

3 LATENT SPACE

The latent space refers to the representation of complex data in a lower-dimensional space by capturing essential features and encoding them into data points. Points that are close to each other within the latent space represent similar or related concepts/features in the original high-dimensional space. For example, in the case of human faces, you could visualize the latent space by systematically interpolating between two points in the latent space and generating images at each step of the interpolation. This will reveal how the model transitions between different facial features, like changing the shape of the eyes, nose, or mouth. While the latent space is part of the AI's 'black box' it is helpful to understand its topology and the way this topology informs generative outcomes of AI tools.

4 INPUTS

AI image generation relies on a range of inputs and parameters. The most common inputs include text prompts called 'text-to-image' and image inputs called 'image-to-image'. In many cases, particularly when creating design variations, the combination of both is used. When using multiple inputs there is an opportunity to weight the impact of individual inputs against each other. With image inputs, assigning the percentage of adherence to the source image provides AI with guidance on how closely synthesized images should resemble the original. This resemblance is defined in the context of the latent space not a human visual reading of the image. Similarly, text prompts can have various strengths. There are additional parameters that define image syntheses, such as generation steps, negative prompts, and seeds discussed later in this paper.

The use of text and image prompts, but particularly text prompts, significantly changes the nature of a creative process. Traditionally, artists and designers were educated to operate in a visual space with heavy utilization of manual skills and capabilities. Crafting with tools—brush, chisel, or just hands—was a prerequisite for artistic expression and freedom. With AI this skillset is changing significantly. Prompting is a new term that allows artists to interact and collaborate with AI systems effectively and direct their behavior



Figure 1: AI-based Japanese building detail

toward specific objectives. Well-designed prompts can help achieve specific goals, control model outputs, and improve the model's performance in desired tasks while poorly chosen prompts can lead to unintended biases or undesired outcomes. Prompting is an art in itself as it requires a good understanding of the artistic vocabulary as well as terms used in model training. This points to another important characteristic of AI generative systems. While model training and imagery used for this training attract the majority of generative attention, creative outputs are conditioned not only by the trained models but also by AI-driven interpretations of prompts.

What is being preserved from the past artistic toolset, and perhaps will continue as the creative constant, is the visual thinking and sophistication required for the evaluation of artistic outcomes, independent of whether they are created by humans or by AI models.

5 VARIATIONS

Not only do AI creative processes involve effective prompting and image inputs but also the ability of an artist to guide versioning of the synthesized outputs over multiple generations. Variations provide an opportunity to transcend from a derivative to transformed imagery with unique visual and conceptual reading. A human role in guiding the evolution of the image can be particularly challenging and rewarding when the original AI models have a limited set of references to prompts used in generating imagery. Figure 1 shows such a condition when developing imagery of plausible Japanese building detail.

However, the variations can also be a nuisance. The difficulty of dealing with variations already appeared early in the digitization process when CAD and image processing software allowed designers to develop alternatives that quickly became a hindrance in the design process. However, this time with AI generative tools, these alternatives require less effort and produce significantly more versions, often with minute amounts of differences. Directing these creative streams and choosing proper outputs will require major human involvement (Figure 2).

6 STYLE TRANSFERS

An artistic style can be one of many characteristics of the higher dimensional image that can be dimensionally reduced and encoded into the latent space. Images associated with the same style would be placed closer to each other while images representing diverse



Figure 2: Subsequent variations often produce minimally differentiated imagery. The ability to curate and select final imagery often may be challenging due to the oversupply fatigue.



Figure 3: A broad range of artistic style outcomes is achieved through balancing of various degrees of style weights. The styles used include those of Zdzislaw Beksinski, Stephen Gammell and Junji Ito. While there are certain similarities to the original inspirations, the synthesized imagery provides transformational visual quality.



Figure 4: Mapping Vienna Secession style of Gustav Klimt into an interior lobby space.

styles would be put further apart within the latent space. Interpolating between these latent space data points allows for mixing multiple styles and other characteristics (Figure 3). This provides generative AI tools with the ability to combine and transfer various image features and characteristics, such as artistic styles, image subjects, and compositions during image synthesis. The interpolation between various points in latent space can also help to transfer features between categories. For example, applying Gustav Klimt's visual quality of his two-dimensional paintings into a three-dimensional architectural space (Figure 4) becomes a transformative factor in developing novel spatial expressions.

7 AUTOMATION BIAS

While human-AI interaction in the context of creativity and design is often seen as a partnership there are cognitive mechanisms that may impact that relationship. One of them is called automation complacency (AC) aka automation bias, which refers to a psychological phenomenon where humans place excessive trust or reliance on automated systems, assuming they are infallible and can make decisions without human intervention [Parasuraman and Manzey, 2010]

In the context of generative AI, AC can occur when blindly accepting AI-generated content or treating it as original work, the quality of which can be easily produced by the general public. AC may also lead users to overlook biases and errors associated with trained models, assuming that the AI is unbiased and accurate.



Figure 5: Mis-constructed imagery may confuse visually similar elements, such as legs with vines (center right), in larger scenes the number of legs may also be incorrect (left), the functional elements such as circular stairs or furniture will privilege visual appearance over usability (right).

7.1 Overcoming Automation Complacency

Direct engagement and control of the creative process could be a form of mitigating a possible AC while understanding the limitations and potential biases of AI systems. Generative AI can effectively assist in creative processes, but human creativity and judgment remain essential in producing meaningful and responsible content. This direct engagement can also help to overcome the loss of personal touch and uniqueness present in many generic generative AI models. The personal touch can be achieved through image inputs and prompt guidance as well as through custom-trained models discussed later.

To foster creativity in the context of AC it is important to consider the following strategies: (1) emphasize, (2) human involvement, (3) experiment and iterate, (4) use generative AI as a starting point, (5) combine multiple AI models or tools, (6) continuously evaluate and critique, (7) set creative challenges, (8) collaborate with others, (9) focus on the learning/creative process

By using these, and similar, strategies creators can better control processes and transcend the generic AI outputs enhancing their artistic expression and fostering a more enriching and fulfilling creative experience.

8 PROCESS CONSIDERATIONS

The following are strategies to successfully engage AI-based creative processes.

8.1 Image fidelity

As generative AI imagery emerges out of the latent space through local emergence, there is no overall and holistic understanding of the synthesized image. This is particularly visible when creating batalistic scenery with multiple horse-mounted warriors, complex human sculptures, or spiral staircases. While these images seem convincing in first sight, a closer look reveals that the number of legs does not correspond to the number of horses or people in the scene. In the case of the round staircases, current AI generative tools mis-construct scenery with steps that do not follow overall perspective or geometric logic. Similarly, furniture design can present functional issues as the primary driver is visual style (Figure 5). This is where human intervention is very helpful in curating variations and making design determinations.

8.1.1 Negative prompts. One of the features of generative AI that partially helps in reducing unwanted image artifacts discussed earlier is a negative prompt. Typically, when interacting with a language model of the AI system, users provide prompts to instruct the model on what information they want it to generate. A regular or positive prompt provides explicit instructions on what the model should include in its response. A negative prompt explicitly instructs the model on what not to include or consider in its output by guiding the model away from generating specific types of content that may be unwanted or inappropriate. However, negative prompts are more effective in eliminating asymmetrical or disfigured faces than accounting for the overall number of items in complex scenery.

8.2 Lateral thinking

Stable diffusion and other generative tools manifest image associations similar to lateral thinking as defined by Edward De Bono [De Bono, 1970] [De Bono, 1995] or the compositional and transition effects that are commonly used in motion graphics when transitioning from one scene to another. Images variations and evolutions within AI focus, or are structured around, shapes and their compositional place within the overall image, not their three-dimensional connotations or the overall meaning of the image. In these cases, they follow more latent space directives than provided prompts (Figure 6). The similarity of AI image generation tools to lateral thinking lies in the ability of AI tools to break down visual patterns and redefine visual space while synthesizing a new image.

8.3 Visual Space Redefinition

AI generative tools like stable diffusion or DALL-E allow for image generation with a combination of a text prompt and an image input. This approach allows artists to take an image input (figure 7, left) and amend it with the text description (prompt). The resulting image is a transformation of the visual space of the image—more 2D than its 3D meaning. In most cases, this approach maintains the overall composition of the 2D image, which also means that the overall 3D perspective view is often preserved, as the images tend to blend from a more transformed center to a less deformed perimeter. However, the level of transformation or fidelity to the original image can be controlled by apportioning the pull strength of the text prompt against the input image.



Figure 6: The latent space facilitates visual drifting between semantically distinct elements that share similar visual characteristics. A waterfall morphs into a white tower structure.

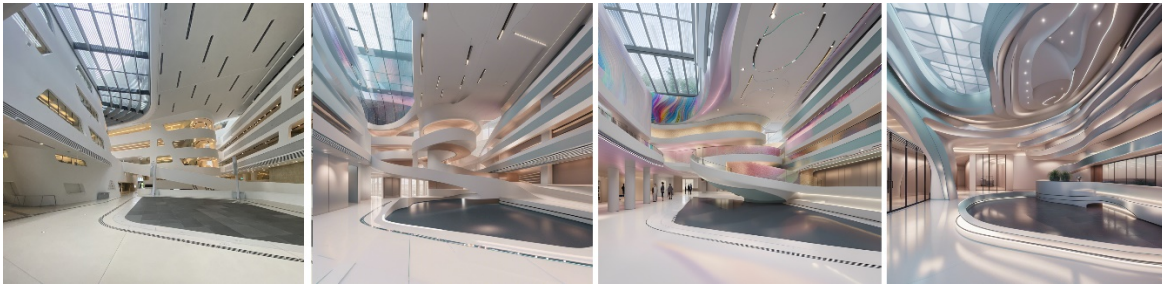


Figure 7: The original photograph of an interior space used as the image-to-image input (left), sublet stages of image transformation aiming at derivative alternatives (two center images), and a significantly transformed image (right).

While the initial transformation usually feels derivative of the original image (Figure 7, two center images), the consequent variations appear to transcend the original precedent. (Figure 7, right) However, both of these scenarios can be very effective in the context of the educational curriculum and design practice. The derivative scenario allows for original image enhancement when the original image is a crude rendering or a change of the visual style is required. This is effective with Adobe Photoshop collages of a rendered image placed in the photographic context that often feels out of place as compared to the computer-generated rough image. In this case, the derivative scenario plays the role of the image post-processor and enhancer. It can also be used for creating variations in the search for the optimal design solution.

Another way of following this process is to go through a series of variations, using the same or slightly modified prompt, to allow for the natural evolution of the visual space of the image. At that point, the final image usually acquires transformation quality so they are no longer perceived as a derivative of the original image.

8.4 Expanding visual image space

There are two related concepts in AI image generations that follow similar creative logic as the previous example: (1) out-painting and (2) in-painting. The out-painting allows for the expansion of the image space. This can be done automatically or manually using the same or modified prompt.

The major benefit of using this tool is the ability of AI to reconcile the edge condition—a natural transition from the original image into a newly created part. This transition provides not only visual integrity of the image but also conceptual as the relationship

between image pixels and image prompts is considered (Figure 8). Even with the same image prompt, it seems that the entire original image is considered not just neighboring pixels as one would do manually in image editing software (Figure 9). In addition to standard image manipulations, this technique also allows for the development of panoramic images (scenery) and in some instances merging context information with the original image.

The in-painting follows a similar methodology by allowing for the substitution of the parts of the original image with a new text-to-image generated section. This approach provides a convenient toolset for image doctoring and refinement. One could remove problematic parts of the original image or add elements to the scenery, such as people and furniture into the interior space. Similarly, the in-painting technique maintains the integrity of the image both on the immediate-boundary pixel level as well as on the broader conceptual consistency, assuming that the prompt allows for this.

8.5 Combining unrelated ideas

Clashing unconnected, or even contradictory, concepts is a good way to step outside known precedents and established creative processes. This approach was heavily utilized by artists in the past, particularly in Surrealist art where tea caps were layered with animal fur (lunch in fur), watches were melting, or locomotives were of the scale of the fireplace. These unexpected assemblies and semantic combinations proved to be fruitful ways to break with stereotypes and generic thinking. To some extent, they also relate to the creative approach discussed by [De Bono, 1970] [De Bono, 1995] as lateral thinking. Similarly, with current AI tools, we can effectively use these techniques. This can be executed both on the

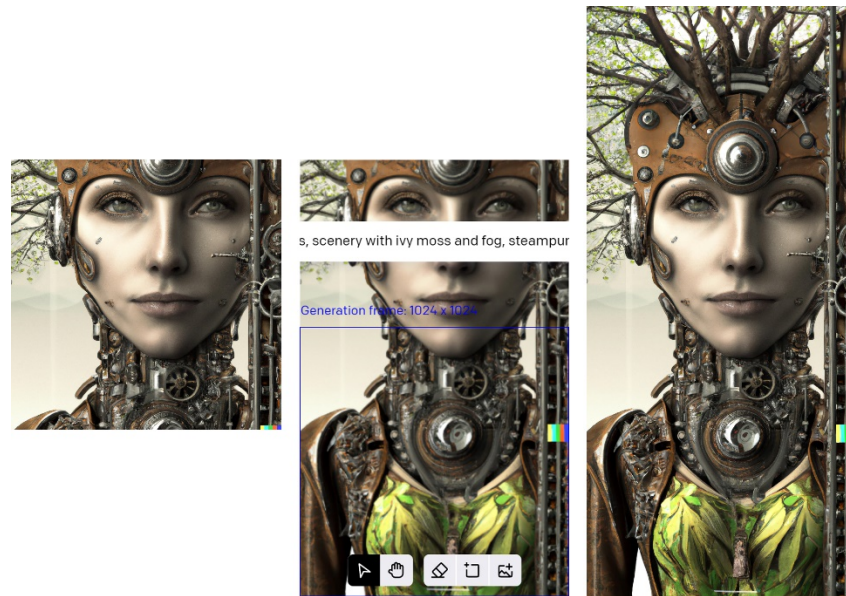


Figure 8: AI-generated out-painting with DALL-E; the original AI image created with a text prompt (left), a screenshot of the process of out-painting with the area below the original image selected for expansion (center), and the final image with two sets of out-paintings one on the bottom and one top of the original image (right).



Figure 9: Automated AI-generated out-painting of the bottom of the image not only extends the original image concept but also responds to a sophisticated condition with image reflection in the wet sidewalk. The source image for the out-painting was the upper square.

level of objects when blending multiple aesthetics and functions as well as when mixing various visual or design styles as when combining multiple artists, or working with a single artistic style and using prompts to expand original artist's applications/aesthetics (Figure 10). This is possible due to the nature of the latent space, discussed earlier, where various concepts or imagery are reduced to a point in a vector space. Interpolation between these points parallels the interpolation between ideas, visuals, and conceptual constructs.

8.5.1 Reverse engineering of the latent space. While cross-blending discrete characteristics to achieve new qualitative outputs transforming original references is a primary goal of the generative use of AI tools. There are also conceptual applications when a certain aesthetics or visual style can be seen as an interpolation of distinct concepts as points in latent space. For instance, combining prompts for baroque architecture style with the style of the industrial revolution results in steampunk-like imagery. If this condition was consistent across a larger set of terms, the latent space could be used to reverse map various visual sets and organize them as a broader cultural construct. Huang et al., 2021 use indirect explorations of the latent space by training a model with 'alpine' architectural characteristics and later exploring design possibilities emerging out of the model. The authors demonstrate the effectiveness of this approach for architecture, especially for conceptual design. Similarly, [Newton, 2019 explores the analytical abilities of AI models to gain insight into architectural styles and cultures.

8.6 Compositions

AI-synthesized imagery retains many characteristics of the original images. Composition and perspectival arrangements are some of these qualities. While AI-synthesized images appear to be well composed producing visually-attractive outputs, this may also be a creative limitation of the AI model as images stay with the same compositional genre. For example, interior spaces may be crafted with different materials and lighting conditions but they may maintain the same perspectival composition. In this situation, the generative aspects of AI were limited to expressions and materiality of interior surfaces not necessarily spatial tectonics.



Figure 10: AI imagery that blends sculptural styles inspired by Magdalena Abakanowicz, an artist working with weaved objects, into leather-like or cardboard-fabric-like wrapped forms

8.7 Image Consistency

A significant challenge with currently available generative AI tools is the consistency of the image outputs. So far, the paper focused on the variations and iterations as creative tools in the pursuit of innovative imagery. In the production stage, particularly in narrative arts, such as storytelling, gaming, and motion graphics, there is a need for consistency of characters and scenery. A character not only needs to look the same (or at least similar) in various sections of an illustrated story but also it needs to resemble itself in various poses and viewing angles. While this is not always achievable there are several shortcuts used by artists to get desirable outcomes. As discussed earlier text-to-image and image-to-image techniques are definitely helpful in getting the general consistency of the visual space but it is not satisfactory when working with character designs or more specific scenery. In this situation, the best current tool is to use the same seed value for all images in a series in addition to other considerations.

In generative AI, a "seed" refers to a fixed or random input that is used to initialize the generation process. The seed plays a crucial role in consistent and reproducible outputs. It acts as a starting point for generating data, such as images, text, or music. When the same seed is used, the generative model will produce the same or very similar outputs allowing for deterministic behavior. This is particularly useful in scenarios discussed earlier, such as visual narratives, to ensure consistent results.

Consequently, the difference between design variations (probabilistic outputs) and design consistent aesthetics (deterministic outputs) lies in the effective use of seeds enhanced with proper prompting and image inputs as secondary drives.

8.8 Custom Models

While an initial engagement with generative AI tools involves default AI models, a significant improvement in the quality and novelty of work can be achieved by using a wide range of models, each trained on a different dataset. Using personalized training sets for generative AI offers several benefits that can enhance the performance, visual diversity, and fidelity (prompt-to-image fitness of generated outputs. Training your own model allows a human

artist to control the sample set used to train the model and, thus, define its style and prompt connotations (Figure 11).

9 DISCUSSION

The focus on the process discussed in this paper does move away from the art creation from concept-based and preoccupation with the initial intent to methodology-based with elements of discovery—revealing and unearthing the class of possible art solutions. With AI and the 'unbearable' effortless generation of the multiplicity of outcomes self-curation of work becomes essential. This is why preventing automation complacency should be integrated into creative methodologies. The ability to navigate this meta landscape with many parallel and alike solutions poses the second creative challenge that requires the synthetic ability to reduce the vastness of outputs and direct them into meaningful and optimal solutions.

Prompt-based creation is a significant change in the tradition of art and design. It privileges verbal over visual or manual skills in the initial phases of the design generation. While manual tool mastery may no longer be required, visual judgment and thinking still apply in the synthetic part of the process when a multiplicity of design alternatives need to be evaluated and synthesized. This is not a singular moment in the process of art creation or discovery but rather a continuous validation process triggered by each set of newly produced AI-synthesized images.

AI generative tools provide a unique opportunity to pursue variations of a particular art instance controlled by the degree of similarity (adherence to the original) and the steps in the AI process that contribute to the refinement. These variations, not unlike in nature, provide additional degrees of separation from the original and allow the artwork to drift (evolve) through the latent space of neural networks.

Through a selective use of prompts, parameters, and controlling variations the intentionality of an artist can be maintained to a certain, if not high level. While generative AI systems produce highly imaginative outcomes, human intervention with a deep technological understanding drives artistic intentionality and agency. This does not preclude future AI systems to be fully autonomous, which would have to include a high level of aesthetic evaluation abilities. However, the current implementations of generative AI



Figure 11: Different models produce vastly different outputs even with the similar prompt and image inputs, and corresponding parameters.

demonstrate wide variations of outputs from high- to poor-quality images. There is no effective system that would be able to validate output visual quality.

10 CONCLUSIONS

This paper responds to the unfolding debate on the use of generative AI tools in art and design. As the conversations and technological frameworks remain in their early stages the paper focuses on emerging possibilities and methodologies rather than on broader claims or conclusions. Thus, the discussion is focused on ways to engage AI creativity and to enter the discussion in the context of practice (artists and designers) and education. The discussed methods can serve as an effective entry point for education and curricula developments with practical advice and conceptual framing to contextualize emerging AI tools in a broader artistic culture and legacy.

While the discussion on autonomous creativity is a significant part of the future of AI system developments and their relationship with humanity, this paper specifically limits its contribution to

human-to-AI interactions as an important step in understanding intentionality, agency, and identity in that newly formed landscape.

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